High-Performance Optimization for Cased-Hole Robot Fusion Positioning Simulation

forename surname[1*]

[1*]Department, University, City, State, China.

Corresponding author(s). E-mail(s): author@example.com;

Abstract

Cased-hole traction robots require precise positioning in GPS-denied, magnetically-interfered downhole environments. The SINS/dual-odometer/EKF fusion positioning simulation involves numerous sampling points with intensive sequential matrix operations, where runtime overhead from memory allocation, full-sequence search, and per-step matrix reconstruction becomes a bottleneck for validation and parameter tuning. This paper proposes a high-performance optimization method for the fusion positioning simulation. Dynamic memory expansion is removed through pre-allocation, full-sequence obstacle search is replaced by O(1) index mapping, and numerical computation is decoupled from visualization and I/O. Reusable state-transition and observation matrix templates, incremental block updates, and inlined high-frequency quaternion/DCM operations reduce per-step EKF computation by approximately 70%. Innovation-based adaptive Q/R tuning, UD covariance propagation, and adaptive odometer anomaly thresholds are further introduced to improve numerical stability and positioning robustness. Experimental results demonstrate significant speedup while maintaining positioning RMSE within an acceptable range compared to the baseline.

Keywords: Cased-hole robot, Fusion positioning, Extended Kalman filter, High-performance optimization, SINS/odometer integration, Adaptive filtering

1 Introduction

1.1 Problem Statement

In horizontal and extended-reach well logging operations, cased-hole traction robots must perform autonomous traversal, instrument delivery, and station-keeping in extreme downhole environments characterized by the absence of GPS signals, unreliable geomagnetic references due to metallic casing interference, high temperature and pressure, and smooth pipe walls prone to slippage.

Precise, stable, and reliable positioning and pose estimation form the foundation for autonomous navigation, path planning, collision avoidance, and high-quality logging operations. Metallic casings create strong magnetic interference that renders magnetometers and compasses ineffective, while individual sensors suffer from noise, bias drift, slippage, and accumulated errors that cannot satisfy long-distance downhole positioning requirements.

Consequently, multi-sensor fusion combining IMU and wheel odometers has emerged as a critical technical approach for cased-hole robot positioning. IMU provides short-term continuous motion estimation but SINS errors accumulate over time; odometers offer velocity information without drift but are susceptible to slippage. EKF-based integrated positioning achieves complementary fusion of both sensor modalities.

The existing simulation system covers 500 m long-distance trajectory generation, random obstacle perturbations (couplings, corrosion, scaling), IMU/dual-odometer data simulation, SINS mechanization, and 15-state EKF fusion positioning, with data scales reaching 166,668 sampling points. This system exhibits characteristics of long-sequence recursion, small-matrix intensive computation, and multi-stage pipeline processing.

As simulation distance, sampling frequency, experiment batch count, and parameter search scale increase, runtime, memory access, matrix construction, and EKF update overhead progressively become key bottlenecks limiting algorithm validation efficiency and engineering iteration speed.

1.2 Challenges

The trajectory simulation stage suffers from dynamic array expansion during obstacle generation and full-sequence traversal search for each obstacle across 166,668 points.

The fusion positioning stage requires the EKF to reconstruct the complete state transition matrix at each sampling period, executing quaternion attitude updates, direction cosine matrix conversions, velocity-position recursion, odometer anomaly detection, filter prediction-update, and error compensation, with computational cost growing linearly with sampling point count.

At the algorithm parameter level, fixed process noise Q, measurement noise R, and anomaly detection thresholds cannot adapt to varying noise levels across different pipe segments, while long-sequence covariance propagation may accumulate numerical errors.

Architecturally, numerical computation is coupled with visualization rendering and file I/O within the same workflow, imposing additional overhead on performance testing and batch experiments.

1.3 Contributions

This paper makes the following concrete contributions:

- We profile the SINS/EKF fusion positioning simulation and remove redundant memory allocation and full-sequence obstacle search through pre-allocation, direct position-index mapping, and computation-visualization-I/O decoupling.
- We reduce the cost of EKF recursion by reusing state-transition and observation matrix templates, incrementally updating only time-varying blocks, and inlining high-frequency quaternion, DCM, and skew-symmetric operations.
- We improve fusion robustness by combining innovation-based Q/R adjustment, UD covariance propagation, and adaptive odometer anomaly thresholds, enabling stable long-sequence positioning while preserving baseline accuracy.

2 Background

2.1 Cased-Hole Robot Positioning

2.1.1 Downhole Environment

The cased-hole robot operates within a sealed metallic pipeline where external absolute positioning information is unavailable. Motion is primarily constrained by axial advancement along the casing and radial clearance limitations. Strong magnetic interference from the metallic casing renders geomagnetic sensors inoperative. Pipe wall defects—couplings, corrosion, and scaling—introduce radial perturbations and slippage risks.

The simulation adopts a casing inner diameter of 130 mm, robot outer diameter of 80 mm, drive wheel diameter of 15 mm, and single-side radial clearance of 25 mm.

2.2 Sensor and Fusion Model

2.2.1 Sensors and Coordinates

The sensor suite comprises a three-axis gyroscope, three-axis accelerometer (IMU), and dual-path odometers. Four coordinate systems are defined: the IMU frame (coincident with the robot body frame), the odometer frame, and the navigation frame (local-level frame with x-axis along the forward direction, y-axis horizontal-right, z-axis vertical-up).

2.2.2 Fusion Model

The positioning scheme employs SINS as the primary navigation system, dual odometers as auxiliary sensors, and EKF as the fusion framework. The system workflow proceeds through: initial alignment → SINS mechanization (quaternion attitude update, velocity recursion, position recursion) → odometer assistance (dual-path averaged displacement increment) → anomaly detection (SINS vs. odometer displacement increment comparison) → EKF error estimation and compensation.

The 15-dimensional error state vector comprises position errors, velocity errors, misalignment angles, gyroscope biases, and accelerometer biases:

$$\mathbf{x} = [\delta\mathbf{p}^T, \delta\mathbf{v}^T, \boldsymbol{\phi}^T, \boldsymbol{\varepsilon}^T, \nabla^T]^T \in \mathbb{R}^{15} \tag{1}$$

The observation model switches between normal mode (3-dimensional: displacement difference + Y/Z velocity constraints) and abnormal mode (2-dimensional: radial velocity constraints).

2.3 Simulation Dataset

2.3.1 Trajectory Design

The simulation constructs a 500 m trajectory at 100 Hz sampling rate with 0.3 m/s constant forward velocity, yielding 166,668 sampling points. Obstacles (scaling, corrosion, couplings) are randomly generated every 20–40 m starting from 50 m, with heights of 1–2 mm and Gaussian-profile influence range of ±0.5 m. Differential radial perturbations are applied: couplings (positive Z), corrosion (negative Z), and scaling (Y+Z combination).

2.3.2 Noise Model

Gyroscope noise standard deviation is 0.005 rad/s, accelerometer noise is 0.01 m/s^2, and odometer noise is 0.02 m/s. The output dataset contains 15 structured fields: timestamp, 3D position truth, 3D velocity truth, 3-axis angular velocity, 3-axis acceleration, and dual odometer velocities.

2.3.3 Baseline Accuracy

The fusion trajectory closely matches the truth value, with radial position fluctuations within ±20 mm. Baseline accuracy metrics are summarized in Table 1.

Table 1 Baseline positioning accuracy (500 m trajectory)

Metric	X-axis	Y-axis	Z-axis
RMSE (mm)	1113.11	4.95	8.06
Relative accuracy	0.24%	—	—
Closure error (mm)	1211.83	—	—

2.4 Computational Characteristics

The simulation system exhibits six characteristic computational patterns:

- **Vector generation**: Time series, axial trajectory, radial perturbations, and sensor noise are amenable to batch vectorized generation.
- **Local perturbation**: Obstacles affect only local trajectory intervals, suitable for interval-based indexing.
- **Conditional branching**: Obstacle type determination and normal/abnormal observation switching form branch-intensive loop structures.

- **Small-matrix intensive**: Quaternion updates (4×4), attitude transformations (3×3), and EKF covariance propagation (15×15).
- **Sequential dependency**: Attitude, velocity, position, and EKF states depend on previous time steps, precluding simple temporal parallelization.
- **Visualization and I/O**: Data file read/write and plotting are suitable for decoupling from core computation.

3 Method

3.1 Bottleneck Analysis

Function-level and code-segment-level timing tools are employed to quantify per-stage execution time. The EKF main loop is instrumented with segmented timing to identify the proportion of time consumed by attitude update, state transition matrix construction, covariance prediction, Kalman gain computation, and error compensation. Memory allocation patterns, dynamic array expansion frequency, and per-loop temporary variable construction overhead are analyzed.

3.1.1 Dynamic Memory Expansion and Full-Sequence Search

The obstacle generation stage concatenates arrays incrementally, causing frequent memory reallocation. Each obstacle requires full-sequence traversal across 166,668 points to identify matching position intervals. As obstacle density increases, this overhead grows proportionally.

3.1.2 Repetitive Matrix Construction

Each time step reconstructs the complete 15×15 state transition matrix F, including specific-force-related subblocks and attitude-related subblocks. High-frequency operations—quaternion-to-DCM conversion, quaternion multiplication, and skew-symmetric matrix construction—are invoked 4–5 times per step. Over 166,668 steps, cumulative function call overhead becomes substantial. The observation matrix H is repeatedly allocated and fully assigned each step.

3.1.3 Fixed Parameters and Numerical Stability

Fixed noise parameters Q and R cannot adapt to different noise levels between quiescent and obstacle-perturbed pipe segments. Standard-form covariance propagation may accumulate numerical asymmetry and positive-definiteness loss over long sequences. A fixed anomaly detection threshold may produce misjudgments under varying operating conditions.

The bottlenecks above motivate three concrete optimization directions: reducing redundant trajectory-generation work, lowering the per-step cost of EKF recursion, and improving filter robustness during long-sequence propagation.

3.2 Trajectory Generation and Dataflow Optimization

3.2.1 Memory Pre-allocation and Dynamic Expansion Elimination

For the obstacle generation stage with incremental array concatenation, a global pre-allocation strategy is established: the maximum obstacle count is estimated and storage is allocated in a single operation. The state transition matrix adopts template pre-construction—allocated once outside the loop with only incremental updates inside. Trajectory arrays, state matrices, and intermediate variables are uniformly pre-allocated, eliminating runtime dynamic memory overhead.

3.2.2 Index Mapping Replacing Full-Sequence Search

Leveraging the linear relationship of constant-velocity motion, a position-to-time-index direct mapping is established with $O(1)$ complexity:

$$\mathrm{idx} = \mathrm{round}\left(\frac{x_{\mathrm{obs}}}{v_x \cdot \Delta t}\right) \tag{2}$$

replacing the original $O(N)$ full-sequence traversal search. Under high-density obstacle scenarios (every 2–4 m), this demonstrates significant scalability advantages.

3.2.3 Computation-Visualization-I/O Three-Stage Decoupling

The coupled numerical computation, plot rendering, and file read/write operations are restructured into independent modules. This supports zero-visualization-overhead execution during performance testing and provides clean computational interfaces for batch experiments and parameter sweeps.

3.3 EKF Loop Acceleration

3.3.1 State Transition Matrix Template and Incremental Update

A pre-constructed F template matrix is designed. At each step, only 4 attitude/angular-velocity-dependent 3×3 subblocks require updating:

- Velocity equation: specific-force-related subblock
- Velocity equation: accelerometer-bias-related subblock
- Attitude equation: angular-velocity-related subblock
- Attitude equation: gyroscope-bias-related subblock

This reduces matrix assignment operations by approximately 70%.

3.3.2 Observation Matrix Dual-Template Switching

Fixed templates are pre-constructed for normal mode (3×15 observation matrix) and abnormal mode (2×15 observation matrix) respectively. At each step, only elements related to the current heading angle and quaternion components are updated, avoiding repeated allocation and full assignment of the observation matrix.

3.3.3 High-Frequency Function Inlining and Temporary Variable Elimination

Quaternion-to-DCM conversion is rewritten as in-place inline computation, eliminating function call overhead. Quaternion multiplication is expanded into direct formula computation. Skew-symmetric matrix construction is converted to in-place assignment. This eliminates the 4–5 function calls per step across 166,668 steps, along with associated temporary variable allocations.

3.4 Adaptive Filtering and Stable Covariance Propagation

3.4.1 Innovation-Based Adaptive Q/R Tuning

Traditional approaches use fixed process noise Q and measurement noise R, which cannot adapt to actual noise levels across different pipe segments. We introduce an online adaptive mechanism based on the innovation (residual) sequence: sliding-window statistics of innovation covariance enable real-time adjustment of the Q/R ratio.

This allows the filter to tighten in quiescent segments and relax in perturbed segments, improving overall positioning accuracy. The adaptive rule is:

$$\hat{R}_k = \frac{1}{W} \sum_{i=k-W+1}^{k} \boldsymbol{\nu}_i \boldsymbol{\nu}_i^T - H_k P_k^- H_k^T \tag{3}$$

where $\boldsymbol{\nu}_i$ denotes the innovation at step i and W is the sliding window size.

3.4.2 UD Decomposition Replacing Standard Covariance Propagation

Standard-form covariance prediction $P^- = FPF^T + Q$ and update $P^+ = (I - KH)P^-$ may accumulate numerical asymmetry and positive-definiteness loss over 166,668 recursive steps. UD decomposition (Upper-triangular + Diagonal) rewrites covariance prediction and update:

$$P = UDU^T \tag{4}$$

where U is unit upper-triangular and D is diagonal. This guarantees the covariance matrix remains positive-definite and symmetric, while converting 15×15 matrix multiplications to triangular matrix operations, balancing numerical stability with computational efficiency.

3.4.3 Adaptive Odometer Anomaly Threshold

Traditional approaches use a fixed threshold for anomaly detection, lacking adaptability to varying operating conditions. We introduce an adaptive threshold mechanism based on historical displacement difference statistics (e.g., 3σ criterion):

$$\delta_k = \mu_{\Delta s} + 3\sigma_{\Delta s} \tag{5}$$

where $\mu_{\Delta s}$ and $\sigma_{\Delta s}$ are the running mean and standard deviation of SINS-odometer displacement differences. This enables the system to dynamically distinguish normal fluctuations from actual slippage/spinning based on real-time operating states, reducing false detection rates.

4 Experiments

4.1 Experimental Setup

Experiments are conducted on [CPU, memory, OS specifications]. The baseline version is the original unoptimized simulation program. The optimized version applies the proposed optimization techniques. The dataset comprises a 500 m trajectory at 100 Hz with 166,668 sampling points.

Baseline positioning accuracy (from Table 1): X-axis RMSE 1113.11 mm with 0.24% relative accuracy, Y-axis RMSE 4.95 mm, Z-axis RMSE 8.06 mm.

4.2 Runtime Evaluation

- Trajectory generation timing comparison (pre-allocation + index mapping vs. original).
- EKF per-step timing comparison (template + inlining vs. original).
- Covariance propagation timing comparison (UD decomposition vs. standard form).
- Total single-run simulation time comparison.
- Scalability across trajectory lengths: 100 m, 500 m, 1000 m, 2000 m.
- Scalability across sampling frequencies: 50 Hz, 100 Hz, 200 Hz.
- Scalability across obstacle densities: every 2–4 m vs. every 20–40 m.
- Memory consumption comparison.

4.3 Positioning Accuracy

- Pre/post-optimization X/Y/Z RMSE comparison (ensuring no degradation from baseline).
- Closure error and relative positioning accuracy comparison.
- Obstacle-segment radial perturbation reproduction quality.
- Adaptive Q/R accuracy improvement over fixed parameters.
- Adaptive threshold false detection rate reduction over fixed threshold.
- UD decomposition numerical stability improvement over standard propagation.

4.4 Ablation Study

5 Related Work

5.1 Navigation Simulation Acceleration

Prior work on recursive computation and numerical integration optimization in robotic simulation, real-time computation requirements for multi-sensor fusion positioning

Table 2 Ablation experiment configurations

Configuration	Verification Target
Pre-allocation + index mapping	Speedup from reduced memory expansion and direct obstacle indexing
F/H templates + inlining	Speedup from reusable matrices and high-frequency function inlining
Adaptive Q/R + UD + adaptive threshold	Accuracy and robustness improvement from adaptive filtering and stable propagation
Runtime optimizations combined	Combined speedup from trajectory/dataflow and EKF-loop optimizations
All proposed techniques	Combined performance, stability, and accuracy improvement

simulation, and parallel computation in large-scale Monte Carlo simulation and parameter sweeps provides context for trajectory generation, dataflow decoupling, and EKF-loop acceleration.

5.2 Efficient Scientific Computing

Relevant techniques include vectorization and memory pre-allocation in numerical computing, small-matrix operation optimization and batch matrix computation, and operator fusion and function inlining strategies for computation-intensive loops.

5.3 Adaptive and Stable Filtering

Innovation-based adaptive Kalman filtering methods, UD decomposition and square-root filters for numerical stability improvement, and adaptive thresholds in fault detection and isolation directly inform adaptive fusion and stable covariance propagation.

6 Conclusion

This paper presents a high-performance optimization method for cased-hole robot fusion positioning simulation:

- Pre-allocation, direct position-index mapping, and computation-visualization-I/O decoupling reduce redundant trajectory-generation and dataflow overhead.
- Reusable F/H templates, incremental block updates, and inlined quaternion/DCM/skew-symmetric operations lower the per-step cost of large-scale EKF recursion.
- Innovation-based Q/R adjustment, UD covariance propagation, and adaptive odometer anomaly thresholds improve long-sequence fusion stability and robustness while preserving positioning accuracy.

Acknowledgements. [To be added.]

Declarations

- Funding: [To be added.]
- Conflict of interest: The authors declare no conflict of interest.
- Data availability: Simulation data and code are available upon request.